TERRAFORM ACROSS CLOUDS

One Terraform project, multiple clouds: what can you actually reuse?

Reusable modules are not the same thing as one universal configuration.

✓ Reuse

Workflow: fmt → validate → plan → review → apply

Module interface: same inputs (name, env, cidr, tags)

Standards: naming, tagging, policy checks

CI pipeline and remote-state pattern

✗ Cloud-specific

Resources: aws_vpc ≠ azurerm_virtual_network ≠ google_compute_network

Identity model and permissions

Network semantics: subnet scope differs per cloud

State backend: S3 / Azure Blob / GCS

```
# Same interface, one implementation per cloud
module "network" {
  source = "./modules/network/aws"  # or /azure, /gcp
  name   = "payments"
  env    = "prod"
  cidr   = "10.20.0.0/16"
}
```

Gotcha: a module's **`source`** must be a literal string. You can't switch clouds with a variable, so keep a separate root configuration per cloud.

01 · THE HOOK

"One Terraform codebase for all three clouds" sounds great until your first resource

There is no universal "network" resource.

`aws_vpc`, `azurerm_virtual_network` and `google_compute_network` are different resources with different arguments and different behavior.

Terraform gives you **one language and one workflow**. It doesn't give you one abstraction over every cloud.

02 · THE PROBLEM

Two traps teams fall into

Trap 1: copy-paste per cloud

Three folders that started identical and quietly drifted apart. Naming, tags and validation differ, and nobody knows which copy is "right".

Trap 2: the universal module

One giant "cloud-agnostic" module packed with count = var.cloud == "aws" ? 1 : 0. It's hard to read, hard to test, and nobody wants to touch it.

The fix: standardize the **interface**, not the implementation.

03 · THE ARCHITECTURE

Shared modules, one root per cloud

```
infra/
├── modules/
│   └── network/
│       ├── aws/      # main.tf variables.tf outputs.tf
│       ├── azure/
│       └── gcp/
└── live/                  # roots: one state each
    ├── aws/prod/network/
    ├── azure/prod/network/
    └── gcp/prod/network/
```

→ **modules/**: one implementation per cloud, same inputs and outputs

→ **live/**: small root configs with the backend, provider and module call

The shared interface

Every cloud's `network` module starts with the same `variables.tf`:

```
variable "name" { type = string }
variable "cidr" { type = string }
variable "tags" {
  type    = map(string)
  default = {}
}

variable "env" {
  type = string
  validation {
    condition     = contains(["dev", "staging", "prod"], var.env)
    error_message = "env must be dev, staging or prod."
  }
}
```

Same names, same validation, same outputs (e.g. `network_id`). Callers learn the interface once.

Different implementations underneath

```hcl
# modules/network/aws/main.tf
resource "aws_vpc" "this" {
  cidr_block = var.cidr
  tags       = merge(var.tags, { Name = "${var.name}-${var.env}" })
}

# modules/network/azure/main.tf
resource "azurerm_virtual_network" "this" {
  name                = "${var.name}-${var.env}"
  address_space       = [var.cidr]
  location            = var.location            # Azure-only input
  resource_group_name = var.resource_group_name # Azure-only input
  tags                = var.tags
}
```

> **Note:** each cloud can add its own extra inputs (Azure needs a location and resource group). In GCP, the CIDR belongs to subnets, not the VPC.

04 · THE GOTCHA

You can't pick the cloud with a variable

```
# ✗ Terraform rejects this: source must be a literal string
module "network" {
  source = "./modules/network/${var.cloud}"
}

# ✓ choose the cloud by folder: one root per cloud
module "network" {
  source = "../../../../modules/network/aws"
  name   = "payments"
  env    = "prod"
  cidr   = "10.10.0.0/16"
}
```

→ **Pin providers** in `required_providers` and commit `.terraform.lock.hcl`

→ **Version shared modules** with Git tags, e.g. `?ref=v1.4.0`

05 · THE TRADE-OFFS

Thin per-cloud modules vs one universal module

Per-cloud modules

✓ Readable and testable
✓ Use each cloud's strengths
✗ Some repeated code
✗ Keeping interfaces in sync takes discipline

Universal module

✓ One entry point
✗ Conditionals everywhere
✗ Lowest common denominator
✗ Every change risks every cloud

Rule of thumb: some duplication is cheaper than the wrong abstraction.

06 · THE TAKEAWAY

Multi-cloud Terraform checklist

- ☐ Same workflow everywhere: fmt → validate → plan → review → apply
- ☐ One module interface per capability, one implementation per cloud
- ☐ Input validation on shared variables
- ☐ One root config (and state) per cloud, env and component
- ☐ Providers pinned, lock file committed
- ☐ Shared modules versioned and tagged
- ☐ One naming and tagging standard, enforced in CI

07 · YOUR TURN

Would you use one repository or separate repositories for each cloud?

Tell me your setup in the comments, and what made you choose it.

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